

Throwing Dice

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to calculate the number of ways to get a sum n by throwing dice. Each throw yields an integer between 1 ... 6.

For example, if $n = 10$, some possible ways are $3 + 3 + 4$, $1 + 4 + 1 + 4$ and $1 + 1 + 6 + 1 + 1$.

Input

The only input line contains an integer n .

Output

Print the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^{18}$

Example

Input	Output
8	125